

## Shuangyan Deng

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Dear UBS Recruitment Team,

I am writing to apply for the **2027 Graduate Talent Program – Group Technology Office – Singapore (Job Reference: 340741BR)**. As a Ph.D. candidate in Statistics at the University of Auckland with research expertise in financial artificial intelligence, large language models, and multi-agent systems, together with previous experience developing data and technology solutions at Industrial Bank, I am excited by the opportunity to contribute to UBS as an AI Engineering or Software Engineering Graduate.

My doctoral research focuses on developing reliable AI systems for complex financial reasoning. I have built multimodal benchmarks, agent-based reasoning frameworks, and methods for evaluating and improving AI reliability. My FinMR benchmark for advanced financial reasoning received the **Best Paper Award at ACM ICAIF 2025**, while my subsequent research has explored step-wise evidence acquisition and cross-model error reflection for improving the reliability of multimodal AI systems. Through these projects, I have gained hands-on experience in AI system design, retrieval and evidence grounding, agent coordination, model evaluation, error analysis, and responsible AI development.

This research background aligns particularly well with UBS's emphasis on the experimental and responsible use of AI to improve outcomes and efficiency. The role's focus on developing technical solutions and maintaining high standards across data quality, data flows, governance, and operational systems is especially appealing to me.

Alongside my research experience, I bring practical experience in financial technology and production-oriented data engineering. At **Industrial Bank**, I contributed to an online wealth-management product sales analytics platform and a real-time marketing operations dashboard. I developed data-processing workflows and microservice-based back-end components, integrated APIs, and worked with technologies including Kafka, ClickHouse, Redis, Spring Boot, and real-time visualization systems. These projects required me to translate business requirements into scalable technical solutions while working across business and technology teams. They also gave me practical experience with data quality, system reliability, real-time data processing, and technology delivery within a financial-services environment. This is one of the strongest parts of your existing letter and is very relevant to the Group Technology Office role.

What particularly attracts me to the Graduate Talent Program is the opportunity to develop at the intersection of **finance, AI, software engineering, and large-scale technology systems** within a global financial institution. I value the program's combination of technical training, collaboration, and rotational opportunities, and I would welcome the chance to learn across different areas of UBS while contributing my interdisciplinary experience in statistics, financial mathematics, finance, AI research, and banking technology. UBS's global technology organization and its close alignment with business needs make this an especially compelling environment in which to begin the next stage of my career.

Thank you for considering my application. I would be delighted to discuss how my background in **financial AI, software and data engineering, model evaluation, and banking technology** could contribute to UBS's Group Technology Office in Singapore.

Yours sincerely,

Shuangyan Deng